

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	4 September 2026
Team ID	ServiceNow
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows how a network request flows from the Service Portal through validation, approval, and automated fulfilment, and the ServiceNow components and external integrations involved at each stage.

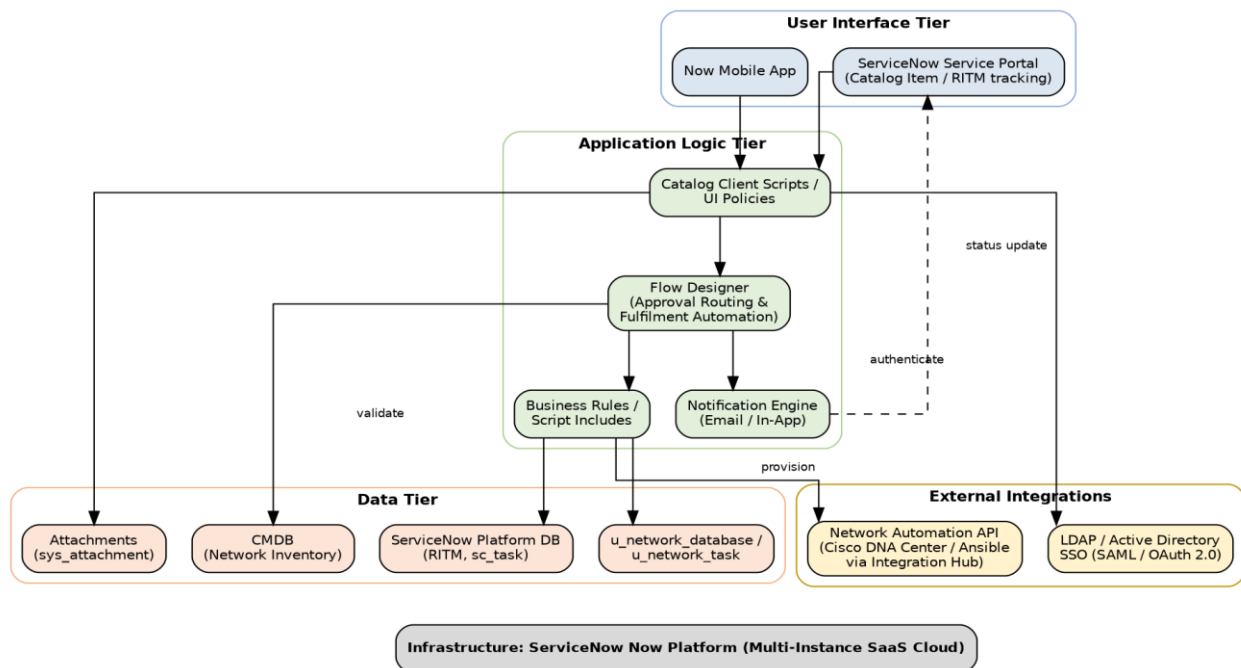


Figure 1: Technical Architecture – Automated Network Request Management (ServiceNow)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the requester and approver interact with the application — request submission, status tracking, and mobile access.	ServiceNow Service Portal (AngularJS widgets, Bootstrap), Now Mobile App
2.	Application Logic-1	Captures and validates catalog item variables on the client side; drives dynamic show/hide behavior of form fields.	Catalog Client Scripts, UI Policies (client-side JavaScript)
3.	Application Logic-2	Orchestrates the end-to-end request lifecycle — approval routing, escalation, and automated fulfillment task creation.	ServiceNow Flow Designer
4.	Application Logic-3	Server-side processing — field validation, policy checks, RITM/task status updates, and reusable functions.	Business Rules, Script Includes (server-side JavaScript / GlideRecord)
5.	Database	Core platform data — request items (RITM), catalog tasks (sc_task), and custom variable/fulfillment tables.	ServiceNow Platform DB, u_network_database, u_network_task
6.	Cloud Database	Configuration and network inventory data used to validate and provision requests.	ServiceNow CMDB (Configuration Management Database)
7.	File Storage	Storage for supporting documents attached to a request (e.g., manager authorization).	ServiceNow Attachments (sys_attachment)
8.	External API-1	Authenticates requesters and approvers against the corporate directory using Single Sign-On.	LDAP / Active Directory Integration, SAML 2.0 / OAuth 2.0 SSO
9.	External API-2	Executes the actual network configuration change once a request is approved.	Network Automation API — Cisco DNA Center / Ansible via ServiceNow Integration Hub
10.	Machine Learning Model	(Optional / future scope) Suggests catalog category and predicts approval likelihood based on historical requests.	ServiceNow Predictive Intelligence
11.	Infrastructure (Server / Cloud)	Application deployment on Local System / Cloud. Local Server Configuration: Not applicable — development done on a Personal Developer Instance (PDI). Cloud Server Configuration: Multi-instance SaaS — ServiceNow-hosted Now Platform (production + sub-production instances).	ServiceNow Now Platform (SaaS / PaaS Cloud)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Front-end widgets on the Service Portal are built using open-source JS/CSS frameworks bundled with the Now Platform.	AngularJS, Bootstrap, jQuery
2.	Security Implementations	Role-based access control on every table, SSO-based authentication, encrypted data in transit/at rest, and secure REST integrations, aligned with OWASP guidelines.	ACLs, SAML 2.0 / OAuth 2.0, TLS 1.2+, Field-level Encryption, MID Server for secure network calls
3.	Scalable Architecture	Multi-tier SaaS architecture where UI, workflow logic (Flow Designer), and data are decoupled, allowing new catalog items, approval chains, and fulfillment flows to be added through configuration.	ServiceNow Now Platform (multi-tier PaaS), Flow Designer
4.	Availability	ServiceNow-managed, geographically redundant data centers with automatic failover ensure the request portal remains available.	ServiceNow High-Availability Cloud Infrastructure (99.9% uptime SLA)
5.	Performance	Platform-level caching of catalog data, indexed custom tables, asynchronous Flow Designer execution for long-running tasks, and CDN-delivered static portal assets keep response times low.	ServiceNow Platform Cache, Asynchronous Flows, CDN

References:

https://www.servicenow.com/docs/bundle/washingtondc-application-development/page/build/service-portal/concept/c_ServicePortal.html
<https://www.servicenow.com/docs/bundle/washingtondc-application-development/page/administer/flow-designer/concept/flow-designer.html>
<https://c4model.com/>
<https://www.servicenow.com/now-platform.html>